Time Series Study Sheet

Based on: Applied Time Series Analysis (Terence C. Mills)

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1 Part 1: Time Series and Their Features

1.1 Fundamental Definitions

A **time series** on a variable x is denoted as x_t, where t is the time index.

- **Observation Period:** $t = 1, 2, \ldots, T$, where T is the last observation.
- **Forecast Horizon:** Future values are denoted as x_{T+h}, where h is the horizon.
- **Sequential Order:** Unlike cross-sectional data, the order of observations in a time series is paramount because of potential temporal dependence.

1.2 Key Features of Time Series

1.2.1 Autocorrelation and Periodic Movements

Autocorrelation refers to the correlation of a series with its own past values. Many series exhibit periodic movements that are not strictly seasonal but repeat over certain intervals.

1.2.2 Seasonality

Seasonality consists of regular, short-term cycles that repeat within a fixed period, typically a year.

1.2.3 Stationarity and Nonstationarity

A process is **stationary** if its statistical properties (mean, variance) remain constant over time.

- **Nonstationarity:** Common in real-world data where the mean or variance changes over time.
- **Integration:** A series is integrated of order d, denoted $I(d)$, if it must be differenced d times to become stationary.
- **First-Difference Operator (Δ):** Used to transform a nonstationary series:

$$\Delta x_t = x_t - x_{t-1} \tag{1}$$

1.2.4 Trends

A trend represents the long-term evolution of a series.

- **Deterministic Trend:** Often modeled as a linear function of time: $x_t = \beta_0 + \beta_1 t + \epsilon_t$.
- **Stochastic Trend:** Represented by a "random walk" where the trend evolves based on cumulative random shocks.

1.2.5 The Concept of the "Level" of a Series

The **level** of a time series represents its local mean or the underlying value at a specific point in time, once noise and seasonal fluctuations are removed.

- **In Stationary Series:** The level is constant and corresponds to the long-run mean μ.
- **In Non-stationary Series:** The level evolves over time. In a Random Walk ($x_t = x_{t-1} + \epsilon_t$), the current level is the sum of all past shocks.
- **State-Space Representation:** In many decomposition models (like Exponential Smoothing), the level L_t is updated recursively:

$$L_t = \alpha x_t + (1 - \alpha)(L_{t-1} + T_{t-1}) \tag{2}$$

  where $0 < \alpha < 1$ is the smoothing parameter, x_t is the current observation, and T_{t-1} is the previous trend estimate.
- **Level Shift:** A structural break where the mean of the series undergoes a permanent jump or drop. This is often modeled using a dummy variable D_t:

$$x_t = \mu + \delta D_t + \epsilon_t, \quad D_t = \begin{cases} 0 & t < \tau \\ 1 & t \geq \tau \end{cases} \tag{3}$$

  where τ is the point of the break and δ is the magnitude of the shift.

1.2.6 Volatility

Volatility refers to a time-varying variance.

- **Volatility Clustering:** Periods of high variation followed by more high variation, common in financial data.
- **Model:** Often handled using GARCH (Generalized Autoregressive Conditional Heteroskedastic) models.

1.3 Natural Constraints

Some series must follow mathematical constraints due to their nature:

- **Count Data:** Observations are non-negative integers $(0, 1, 2 \dots)$.
- **Compositional Data:** Series representing shares of a whole, satisfying:

$$0 \leq x_{i,t} \leq 1 \quad \text{and} \quad \sum_{i=1}^{n} x_{i,t} = 1 \tag{4}$$

1.4 Multivariate Features

- **Common Trends:** Two or more nonstationary series may be "bound together" by a common underlying trend (Cointegration).
- **Causality:** One series may help predict another (Granger Causality).

2 Part 2: Transforming Time Series

Prior to statistical analysis, it is often necessary to transform data to make it more amenable to modeling. There are three general classes of transformations: distributional, stationarity-inducing, and decompositional.

2.1 Distributional Transformations

These aim to achieve approximate normality, symmetry, and constant variance.

2.1.1 Logarithmic Transformation

A popular choice for non-negative series that are skewed to the right. It "straightens out" exponential trends and stabilizes variance.

- **Usage:** Effective when the standard deviation is proportional to the level of the series.
- **Formula:** $y_t = \log(x_t)$.

2.1.2 Box-Cox Transformation

A general class of power transformations that includes the logarithm as a special case for positive x:

$$f_{BC}(x_t; \lambda) = \begin{cases} (x_t^\lambda - 1)/\lambda & \lambda \neq 0 \\ \log x_t & \lambda = 0 \end{cases} \tag{5}$$

2.1.3 Handling Negative Values

For series with negative values, several alternatives exist:

- **Signed Power:** $f_{SP}(x_t; \lambda) = \text{sgn}(x_t)|x_t|^\lambda - 1/\lambda$ for $\lambda > 0$.
- **Inverse Hyperbolic Sine (IHS):** Useful for extreme values of either sign:

$$f_{IHS}(x_t; \lambda) = \frac{\sinh^{-1}(\lambda x_t)}{\lambda} = \frac{\log(\lambda x_t + (\lambda^2 x_t^2 + 1)^{1/2})}{\lambda} \tag{6}$$

2.2 Stationarity Inducing Transformations

To remove deterministic or stochastic trends, differencing is applied.

2.2.1 First Differencing

Used for series that are integrated of order 1 ($I(1)$):

$$\Delta x_t = (1 - B)x_t = x_t - x_{t-1} \tag{7}$$

where B is the backshift operator ($Bx_t = x_{t-1}$).

2.2.2 Seasonal Differencing

For a series with a seasonal cycle of length s (e.g., $s = 12$ for monthly data):

$$\Delta_s x_t = (1 - B^s)x_t = x_t - x_{t-s} \tag{8}$$

2.3 Decomposing a Time Series

Observed series X_t are often decomposed into Trend (T_t), Seasonal (S_t), and Irregular (I_t) components.

2.3.1 Additive vs. Multiplicative Models

- **Additive:** $X_t = T_t + S_t + I_t$.
- **Multiplicative:** $X_t = T_t \times S_t \times I_t$ (often transformed to additive via logs).

2.3.2 Smoothing Transformations (Moving Averages)

A simple way to extract the trend is via a symmetric $2k+1$ point moving average:

$$M_t = \frac{1}{2k+1} \sum_{j=-k}^{k} x_{t+j} \tag{9}$$

2.3.3 Hodrick-Prescott (HP) Filter

A more advanced decomposition that minimizes the following sum for a chosen smoothing parameter λ:

$$\min \sum_{t=1}^{T} (x_t - T_t)^2 + \lambda \sum_{t=2}^{T-1} [(T_{t+1} - T_t) - (T_t - T_{t-1})]^2 \tag{10}$$

Common values for λ include 14,400 for monthly data or 1,600 for quarterly data.

3 Part 3: ARMA Models for Stationary Time Series

3.1 Intuitive Interpretation of AR and MA Components

Before diving into the complex estimations, it is helpful to understand the behavioral "personality" of these models.

3.1.1 AR (Autoregressive): The Long-Term Memory

The **AR** component describes a system that depends on its **own previous values**.

- **Analogy:** It is like a person's mood; if someone was happy yesterday, he is likely to still be somewhat happy today.
- **Mechanism:** Each new value is a weighted percentage of the last value, plus a current "random surprise."
- **Impact:** It creates **momentum** or **persistence**. For example, if the temperature is 30°C today, it is unlikely to drop to 0°C tomorrow because the "memory" of today's heat lingers.

3.1.2 MA (Moving Average): The Shock Absorber

The **MA** component does **not** depend on the previous values of the series, but rather on the **previous random shocks** (surprises or errors).

- **Analogy:** Imagine an earthquake. The "shock" happens today. Tomorrow, we feel an "aftershock." By the third day, that specific shock is gone.
- **Mechanism:** It looks at the "random noise" from yesterday and determines how much of that noise should still ripple into today.
- **Impact:** It handles **short-term echoes**. Once the shock exceeds the order q, it is completely forgotten.

3.1.3 ARMA: The Combined Personality

Combining them into an **ARMA** model allows us to represent a series with:

- **Persistence (AR):** The tendency to stay where it was.
- **Reaction to Shocks (MA):** The tendency to ripple for a short time after a sudden surprise.

3.1.4 Summary Comparison

Feature	AR (Autoregressive)	MA (Moving Average)
Logic	"I am where I am because of where I was."	"I am where I am because of the 'news' I just received."
Memory	Long-term (fades geometrically).	Short-term (cuts off abruptly).
Analogy	**Momentum:** Like a car rolling after we stop hitting the gas.	**Echo:** Like a sound bouncing for a moment, then vanishing.

3.2 Theoretical Foundations

3.2.1 The Stochastic Process

A **time series** is a realization of a **stochastic process** $\{x_t, t \in \mathbb{Z}\}$. We assume the process is **linearly regular**, meaning it can be expressed as a linear combination of its past innovations.

3.2.2 Stationarity (Covariance Stationarity)

A process is weakly stationary if:

1. $E[x_t] = \mu$ (The mean is constant for all t).
2. $Var(x_t) = E[(x_t - \mu)^2] = \sigma^2 = \gamma_0 < \infty$.
3. $Cov(x_t, x_{t-k}) = E[(x_t - \mu)(x_{t-k} - \mu)] = \gamma_k$ for all t.

The **Autocorrelation Function (ACF)** is defined as:

$$\rho_k = \frac{\gamma_k}{\gamma_0} \tag{11}$$

Note that $\rho_0 = 1$ and $|\rho_k| \leq 1$.

3.2.3 The White Noise Process

The building block of all ARMA models is **White Noise** $\{a_t\}$, defined as:

- $E[a_t] = 0$
- $E[a_t^2] = \sigma_a^2$
- $E[a_t a_s] = 0$ for $t \neq s$ (uncorrelated shocks).

3.3 Autoregressive Processes: $AR(p)$

3.3.1 The $AR(1)$ Model: Detailed Derivation

The model is $x_t = \phi x_{t-1} + a_t$.
By repeated substitution (assuming the process started in the infinite past):

$$x_t = a_t + \phi a_{t-1} + \phi^2 a_{t-2} + \cdots = \sum_{j=0}^{\infty} \phi^j a_{t-j} \tag{12}$$

Variance calculation:

$$\gamma_0 = Var\left(\sum_{j=0}^{\infty} \phi^j a_{t-j}\right) = \sigma_a^2 \sum_{j=0}^{\infty} \phi^{2j} = \frac{\sigma_a^2}{1-\phi^2} \tag{13}$$

This requires the **Stationarity Condition**: $|\phi| < 1$.

3.3.2 The General $AR(p)$ and Yule-Walker Equations

The $AR(p)$ model is $x_t = \phi_1 x_{t-1} + \cdots + \phi_p x_{t-p} + a_t$.
To find the autocorrelations, we multiply by x_{t-k} and take expectations. This yields the **Yule-Walker Equations**:

$$\rho_k = \phi_1 \rho_{k-1} + \phi_2 \rho_{k-2} + \cdots + \phi_p \rho_{k-p} \quad \text{for } k > 0 \tag{14}$$

This system of equations allows us to solve for the parameters ϕ_i if we know the autocorrelations ρ_k.

3.4 Moving Average Processes: $MA(q)$

3.4.1 The $MA(1)$ Model

Defined as $x_t = a_t - \theta a_{t-1}$.

- **Mean:** $E[x_t] = E[a_t] - \theta E[a_{t-1}] = 0$.
- **Variance:** $\gamma_0 = E[(a_t - \theta a_{t-1})^2] = \sigma_a^2(1+\theta^2)$.
- **Autocovariance at lag 1:** $\gamma_1 = E[(a_t - \theta a_{t-1})(a_{t-1} - \theta a_{t-2})] = -\theta\sigma_a^2$.
- **ACF:** $\rho_1 = \frac{-\theta}{1+\theta^2}$ and $\rho_k = 0$ for $k > 1$.

3.4.2 Invertibility

An $MA(1)$ is **invertible** if $|\theta| < 1$.
If invertible, we can express the error a_t as an infinite sum of past observations:

$$a_t = \sum_{j=0}^{\infty} \theta^j x_{t-j} \tag{15}$$

3.5 The $ARMA(p,q)$ Process

3.5.1 General Definition

Combining both parts:

$$(1 - \phi_1 B - \cdots - \phi_p B^p)x_t = (1 - \theta_1 B - \cdots - \theta_q B^q)a_t \tag{16}$$

Where B is the **Backshift Operator** ($B^k x_t = x_{t-k}$).
This is often written as $\Phi(B)x_t = \Theta(B)a_t$.

3.5.2 The Partial Autocorrelation Function (PACF)

While the ACF of an AR model tails off, the **PACF** "cuts off." The PACF at lag k (denoted ϕ_{kk}) is the correlation between x_t and x_{t-k} after controlling for the effects of $x_{t-1}, \ldots, x_{t-k+1}$.

- For an $AR(p)$, $\phi_{kk} = 0$ for $k > p$.
- For an $MA(q)$, the PACF tails off (opposite of the ACF).

3.6 Diagnostic Tools: ACF and PACF

To decide whether a series is an AR, an MA, or a mix of both ($ARMA$), we use two primary functions that measure how observations are related to each other over time.

3.6.1 ACF (Autocorrelation Function): The Total Correlation

The **ACF** at lag k, denoted as ρ_k, measures the simple correlation between x_t and its past value x_{t-k}. It is defined as the ratio of the autocovariance to the variance:

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \frac{E[(x_t - \mu)(x_{t-k} - \mu)]}{E[(x_t - \mu)^2]} \tag{17}$$

- **Behavior for MA:** It is the "ID card" for Moving Average models. For an $MA(q)$, the ACF will be significant for q lags and then suddenly drop to zero (**cuts off**).
- **Behavior for AR:** For Autoregressive models, the ACF **tails off** (fades slowly) because the memory of the past persists through a chain reaction.
  For an $AR(1)$, this decay is geometric: $\rho_k = \phi^k$.

3.6.2 PACF (Partial Autocorrelation Function): The Direct Connection

The **PACF** at lag k, denoted as ϕ_{kk}, measures the correlation between x_t and x_{t-k} **after removing the effects** of all intervening observations $(x_{t-1}, \ldots, x_{t-k+1})$. It is formally the last coefficient of an $AR(k)$ regression:

$$x_t = \phi_{k1}x_{t-1} + \phi_{k2}x_{t-2} + \cdots + \phi_{kk}x_{t-k} + a_t \tag{18}$$

- **Logic:** It reveals if x_{t-k} has a *direct* impact on x_t.
- **Behavior for AR:** It is the "ID card" for AR models.
  In an $AR(1)$, the PACF at lag 1 is ϕ, but for all lags $k > 1$, the PACF is 0.
  For an $AR(p)$, the PACF will be significant for p lags and then suddenly **cut off**.
- **Behavior for MA:** For Moving Average models, the PACF **tails off** (fades slowly).

3.6.3 Summary Table for Model Identification

This table is used by analysts to "read" the plots and choose the model order (p, q):

Model	**ACF Plot**	**PACF Plot**
AR(p)	Tails off (geometric decay)	**Cuts off after lag** p
MA(q)	**Cuts off after lag** q	Tails off (geometric decay)
ARMA(p,q)	Tails off	Tails off

3.7 Information Criteria: Choosing the Best Model

When we build a model, we face a trade-off: a complex model fits the current data well but may fail to forecast the future (**over-fitting**), while a simple model might miss important patterns. We use Information Criteria to find the "sweet spot" by minimizing the following:

- **AIC (Akaike Information Criterion):** $AIC = -2\ln(L) + 2k$. It provides a small penalty for each parameter (k). It is often used when the primary goal is high-quality forecasting.

- **BIC (Bayesian Information Criterion):** $BIC = -2\ln(L) + k\ln(T)$. It provides a much larger penalty that increases with the sample size (T). It is "conservative," meaning it prefers simpler models with fewer parameters to ensure the model is truly representative of the underlying process.

Rule of Thumb: Calculate both for various combinations of p and q. The model with the **lowest score** is generally considered the best balance of accuracy and simplicity.

3.8 The Connection: From Identification (ACF/PACF) to Selection (AIC/BIC)

In practice, the process of building an ARMA model follows a specific logical sequence. The Information Criteria (AIC/BIC) act as the final decision-makers when the visual tools are ambiguous.

3.8.1 Step 1: The Detective Work (ACF and PACF)

We use the **ACF** and **PACF** to "shortlist" potential models:

- If the PACF cuts off at lag 2, we suspect an **AR(2)**.
- If the ACF cuts off at lag 1, we suspect an **MA(1)**.
- **The Problem:** In real-world data, the plots are rarely perfect. we might see patterns that look like they could be an $AR(2)$, an $MA(2)$, or an $ARMA(1,1)$ all at once. This is where we need a tie-breaker.

3.8.2 Step 2: The Competition (Estimation)

We estimate all the "suspect" models identified in Step 1. We now have three different versions of reality:

1. Model A: $AR(2)$ (Uses 2 parameters: ϕ_1, ϕ_2)
2. Model B: $MA(2)$ (Uses 2 parameters: θ_1, θ_2)
3. Model C: $ARMA(1,1)$ (Uses 2 parameters: ϕ_1, θ_1)

3.8.3 Step 3: The Final Judge (AIC and BIC)

Since all these models might have a similar number of parameters (k), we use **AIC** and **BIC** to see which one explains the data most efficiently.

- **Why use them with AR/MA?** If add more lags (e.g., jumping from an $AR(2)$ to an $AR(10)$), our model will always fit the training data better, but it becomes "noisy."
- **The Penalty:** AIC and BIC "punish" the model for every extra p (AR part) or q (MA part) we add.
- **Selection:** we calculate the AIC/BIC for all candidates. The one with the **lowest value** is the winner. It tells us exactly how much "memory" (AR) and how many "shocks" (MA) are truly necessary without overcomplicating the model.

3.8.4 Summary of the Relationship

- **PACF/ACF:** Tell us which *direction* to look in (Is it memory-based or shock-based?).
- **AIC/BIC:** Tell us exactly where to *stop* adding complexity (How many lags are enough?).